

Task 1: Data Cleaning Report

I. Introduction

The dataset contained 390 athlete records and 17 variables from the Milano Cortina 2026 Winter Olympics. The variables described athlete performance, physiological characteristics, and country-level factors. Our objective in this task was to prepare a clean dataset for analysis and modelling. To do this, we handled missing values, corrected errors, removed duplicate records, reviewed outliers, and created new features to improve the dataset.

II. Duplicate Records

Duplicate records were initially scanned through the the Athlete_ID values which proved to be insufficient. However, upon scanning the combination of attributes as a super key in the dataset, which included the athlete name and excluded the Athlete_ID, we found there to be 3 duplicate entries. These were removed to bring our total records from 390 to 387.

III. Error Detection and Correction

Thereafter, error detection and correction was the next priority as impossible or erroneously incorrect values may skewer remediation efforts for missing values later. Four impossible/erroneous values were found with the following attributes:

- Age = 9 – while there is no set minimum age for Olympians, this age appears wholly unrealistic given the sport this particular athlete was intended to participate in, ice hockey.
- Training_Hours_Per_Week = 172 – there are only 168 hours in a week assuming this athlete was intending to train 24 hours a day and is therefore impossible.
- Body_Fat_Pct = 99.2 – this value is wholly impossible for an individual.
- VO2max = -8.4 – this value implies that the athlete's body is producing oxygen instead of consuming oxygen during physical activity and is therefore an impossible value.

These above 4 values were then handled/fixed by replacing the former with the mean attribute value within their respective sport (e.g. mean age in ice hockey). This was done to

maintain data integrity without skewing the data in any particular direction. This remediation process carries through to missing data handling.

IV. Missing Data Analysis

Several variables contained missing values with rates ranging from 6% to 26%. The largest affected variable being Reaction_Time_ms at 26%. However, some missing variables were found to be systematic and **missing not at random (MNAR)**. These include World_Cup_Points_Preseason and Reaction_Time_ms. World_Cup_Points_Preseason missing values are indicative of sport specific evaluations that are present in some and entirely missing in others where a comparison across these sports would be inaccurate or impossible. Reaction_Time_ms are also systematically missing values where endurance sports may not need reaction times as a measurable metric in comparison to alpine skiing or ice hockey where these values are imperative to performance analysis. This leaves the remaining missing values which are **missing completely at random (MCAR)**. These variables including training_hours_per_week and altitude_training were deemed to be MCAR given that there appeared to be no systematic reason or pattern unique to a specific group. With respect to missing value handling, MNAR values were left as is, indicating that systematic gaps are not things to be fixed but rather non-considerations to performance within that particular sport. MCAR values however, were adjusted as done with erroneous and impossible values, mean values within each respective sport were used to maintain distributions.

V. Outlier Analysis

We used the inter-quartile range (IQR) and Z-score methods to flag several extreme values. However, not all outliers were errors. These records were reviewed before taking action. For example, Deanna Stellato-Dudek (age 42, figure skating) and Elana Meyers Taylor (age 41, bobsleigh, gold medalist) were flagged as age outliers. These cases are valid. In some sports, experience allows athletes to compete at a high level at older ages. Removing these values would remove real athlete profiles and were therefore left as is.

VI. Feature Engineering

After cleaning the dataset, we created new features to support analysis and modelling. These include the following:

- **VO2max to Body Fat ratio** = $\text{VO2max} / \text{body_fat_pct}$
 - This combines aerobic capacity and body composition.

- **Experience Index** = previous_olympics * training_hours_per_week
 - This reflects both experience and training load.
- **Country Strength** = country_GDP_per_capital * winter_sport_tradition_index
 - This captures the combination of economic and cultural factors to an athlete.

VII. Before vs. After Summary Table

Metrics	Before	After
Rows (records)	390	387
Variables (attributes)	17	24
Unique Athlete IDs	390	387
Missing or Blank Values	310	0

*3 duplicates records removed

*3 attributes created (and 4 measurable binary event attributes)

*All missing and impossible values fixed to be in-line or assigned non-blank values